Equation Sheet: Oscillations & Waves

Intro Physics II • Lectures 1–6 • Knight ch. 15–17

Most sections are split into **fundamental** equations (know these cold — your starting points) and **derived** equations (a line or two of algebra or calculus from the fundamentals).

Symbols

A	amplitude	φ_0	phase constant	λ	wavelength
T	period	k	spring const. *or* wave number	v	wave speed
f	frequency	μ	linear mass density	T_s	string tension
ω	angular frequency				

1. SHM — kinematics (Knight 15.1–15.2)

fundamental

Position (the definition of SHM) — $x(t) = A\cos(\omega t + \varphi_0)$

Angular frequency & frequency — $\omega = 2\pi f, \qquad f = 1/T$

derived

Velocity — $v_x(t) = -\omega A\sin(\omega t + \varphi_0), \qquad v_{\max} = \omega A$

Acceleration — $a_x(t) = -\omega^2 A\cos(\omega t + \varphi_0) = -\omega^2 x$

Period from ω — $T = 2\pi/\omega$

2. SHM — dynamics & energy (Knight 15.3–15.4)

fundamental

Restoring force (Hooke's law) — $F_x = -kx$

Angular frequency, mass on a spring — $\omega = \sqrt{k/m}$

Total mechanical energy (constant) — $E = \frac{1}{2}kA^2$

derived

Period, mass on a spring — $T = 2\pi\sqrt{m/k}$

Energy at any instant — $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \frac{1}{2}m\,v_{\max}^2$

Speed as a function of position — $v(x) = \pm\,\omega\sqrt{A^2 - x^2}$

3. Vertical oscillations & the pendulum (Knight 15.5–15.6)

fundamental

Simple pendulum (small angle) — $\omega = \sqrt{g/L}$

Hanging spring — still SHM, same $\omega = \sqrt{k/m}\,$; equilibrium shifts by mg/k

derived

Pendulum period — $T = 2\pi\sqrt{L/g}$

4. Traveling waves (Knight 16.1–16.3)

fundamental

Wave (sinusoidal)	$y(x,t) = A\sin(kx - \omega t + \varphi_0)$
	$kx - \omega t$: moves $+x$ $\qquad kx + \omega t$: moves $-x$
Wave number	$k = 2\pi/\lambda$
Wave speed	$v = \lambda f$

derived

Wave speed — equivalent forms	$v = \lambda/T = \omega/k$
Transverse speed of a string element	$v_{y,\max} = \omega A$ (*not* the wave speed v)

5. Waves on a string & in pipes (Knight 17.1–17.2)

fundamental

Wave speed on a string	$v = \sqrt{T_s/\mu}\,, \qquad \mu = m/L$
Standing wavelengths — ends alike (both fixed, or pipe open–open / closed–closed)	$\lambda_n = 2L/n\,, \qquad n = 1, 2, 3, \ldots$
Standing wavelengths — one node end, one antinode end (string fixed–free, pipe open–closed)	$\lambda_n = 4L/n\,, \qquad n = 1, 3, 5, \ldots$

derived

Harmonic frequencies	$f_n = \dfrac{v}{\lambda_n} = \dfrac{nv}{2L} = nf_1 \quad \left(\text{or } \dfrac{nv}{4L},\ n \text{ odd}\right)$
Node-to-node (or antinode) spacing	$\Delta x = \lambda/2$

6. Superposition & interference (Knight 16.3, 17.3–17.5)

Principle of superposition	$y(x,t) = y_1(x,t) + y_2(x,t)$
Two in-phase sources — constructive	$\Delta r = m\lambda, \qquad m = 0, 1, 2, \ldots$
Two in-phase sources — destructive	$\Delta r = \left(m + \frac{1}{2}\right)\lambda$

Companion simulations at `hoppese.github.io/Intro-physics-sims/`: `energy-in-shm`, `pendulum`, `wave-explorer`, `standing-waves`, `two-source-interference`.